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Adaptive Meta-Goal Formulation for Long-Horizon Autonomous Research Workflows

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ABSTRACT

Executing end-to-end scientific investigations requires autonomous agents to formulate, decompose, and iteratively revise meta-goals across months of experimentation. We propose an adaptive goal-formulation architecture that combines curiosity-driven exploration metrics with literature gap indicators. The system monitors intermediate experimental divergences, triggering automated hypothesis reformulations when null results occur. In simulated scientific testbeds, our agent independently discovered and validated 12 novel molecular candidates with zero human intervention. The unresolved challenge is managing error propagation across multi-week planning horizons, where early misclassifications compound downstream.

Keywords— Artificial Intelligence, Meta-Learning, Long-Horizon Planning, Autonomous Research

INTRODUCTION

Long-horizon scientific exploration requires autonomous agents capable of persevering and pivoting when hypotheses fail. Current agent frameworks lack mechanisms for recursive meta-goal revision over extended execution spans.

METHODOLOGY

The agent combines Bayesian uncertainty estimation with dynamic literature graph tracking. When experimental uncertainty exceeds empirical thresholds, the planner triggers meta-goal reformulation routines.

RESULTS

The system completed 84 autonomous research cycles, recovering from early negative experimental results in 89% of cases through adaptive hypothesis steering.

LIMITATIONS

Cascading errors remain a vulnerability; an erroneous data filtering step in early phases can bias subsequent hypothesis formulation for multiple days before corrective mechanisms trigger.

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